UPES

Mid-Semester Examination, March 2026

Programme Name:	BTech CSE	Semester:	02
Course Name	: Python Programming	Time	: 02 hrs
Course Code	: CSEG1021	Max. Marks:	100
Nos. of page(s)	: 2	Calculator allowed:	No

Instructions: Please attempt according to the time provided and given weightage.

SECTION A (30 Marks) 6 Questions
Attempt all questions (Not More than 50 words)

S. No.		Marks	CO
Q 1	What is the purpose of the break and pass statement in Python? Explain its usage with the help of a code snippet.	6	CO1
Q 2	List the mutable and immutable data types in Python. State the major differences with relevant examples.	6	CO1

Q 3	Predict the output and explain it
	`data = {"a": 1, "b": 2, "c": 3}`
	`print(data)`
	`for key in list(data.keys()):`
	`    print(f"{key}")`
	`    if data[key] % 2 == 0:`
	`        data[key] += 5`
	`        print(f"{data[key]=}")`
	`    else:`
	`        data[key] = data[key] * 2`
	`        print(f"{data[key]=}")`
	`print(data)`
	|a;
	a:1, b:2, c:3
	a
	a=2
	b
	b=7
	c
	c=6
	a:2, b:7, c=6
		6	CO2

Q 4	Differentiate between any four Python collections based on properties such as ordering, mutability, and duplication, with suitable examples.	6	CO2
Q 5	Mention the use of various file access modes (w, a, w+) in the suitable file handling function in Python.	6	CO3

SECTION B
(45 Marks) 3 questions – (Q8 should have internal choice)

Q 6	a) Discuss the conditions to check if a year is a leap year. (5 Marks) b) Justify, Is the year 2000 a leap year? Is 1900 a leap year? Justify your answer logically. (5 Marks) c) Develop an application which says the entered year is a leap year or not. (5 Marks)	15	CO1
Q 7	a) Develop an Electricity Bill Calculator using the **conditional statements** with the following conditions specified below by getting the units consumed.(10 Marks) • For the first 50 units – Rs. 3.50/unit • For next 100 units – Rs. 4.00/unit • For next 100 units – Rs. 5.20/unit • For units above 250 – Rs. 6.50/unit	15	CO1

	b) Evaluate the following expressions and justify your answer based on operator precedence rules: (5 Marks) 10 + 5 * 2 > 25 or 3 * 3 == 9 and not False		
Q 8	The State Urban Development Department is maintaining a record of major cities in a file named city.txt. The file contains details of five cities in the following format: **CityName Population(in lakhs) Area(in sq. km)** **Dehradun 5.78 308.20** **Delhi 190 1484** The department wants to analyze the data to support planning and resource allocation decisions. Based on the above case, write a Python program to perform the following tasks: a) Read the file and display the complete details of all cities.(5 Marks) b) Identify and display the names of cities having a population greater than 10 lakhs.(5 Marks) c) Calculate and display the total area covered by all cities listed in the file.(5 Marks) **OR** Write a Python program to store a set of integers in a file. Read the integers from the file and perform the following: a) Find the maximum number. (5 Marks) b) Calculate the average of all numbers. (5 Marks) c) Count how many numbers are greater than 100. (5 Marks)	15	CO3
	SECTION C **(25 Marks) 1 Questions (Q 9 should have internal choice)**		
Q 9	You are working as a Python Developer in a small software company. The company wants to develop a simple Contact Book Application to help users store and manage their personal contacts efficiently with the following functionality. a) Identify the data structure that suits well to the problem and justify.(5 Marks) b) Add new contact (Name and Phone Number) (5 Marks) c) Search contact by name (5 Marks) d) Update contact number (5 Marks) e) Delete contact (5 Marks) **OR** a) Differentiate between Positional parameters, Keyword parameters, Optional parameters, and Default arguments in Python with suitable examples. (10 Marks) b) Write a Python program to calculate the volume of multiple cones stored as (radius, height) pairs in a list. Hint: cones = [(3, 5), (4, 6), (5, 10)] and $V = \frac{1}{3}\pi r^2 h$ I. First, implement it using a normal function.(5 Marks) II. Then rewrite it using a lambda function inside map().(5 Marks) III. Justify which approach is more suitable in this given scenario.(5 Marks)	25	CO2